

# Measurements

Phys 512

Ch – 6, 7,8 David

## Time Evolution

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = H\psi \quad (6.2)$$

Time evolution of quantum state

$$|\psi(t)\rangle = e^{-iHt/\hbar} |\psi(0)\rangle \quad (6.3)$$

$$U = e^{-iHt/\hbar} \quad (6.4)$$

Time evolution of density operator

$$\rho_t = U\rho_0 U^\dagger \quad (6.5)$$

$$\text{if } \text{Tr}(\rho) = 1 \quad \Rightarrow \quad \text{Tr}(\rho_t) = 1$$

Dynamics of quantum system  
is Trace preserving

## Measurements

### PROJECTIVE MEASUREMENTS

$$P = P^\dagger \quad (6.6)$$

$$P^2 = P \quad (6.7)$$

Two projection operators,  $P_1$  and  $P_2$  are orthogonal if their product is zero, i.e. for every state  $|\psi\rangle$

$$P_1 P_2 |\psi\rangle = 0$$

$$P_i P_j = \delta_{ij} P_i \quad (6.13)$$

A complete set of orthogonal projection operators is one for which

$$\sum_i P_i = I \quad (6.9)$$

$$\sum_i |u_i\rangle \langle u_i| = I \quad \text{completeness}$$

If **two projection operators commute**, then their product  $P_1 P_2$  is also a projection operator

$$\text{Pr}(i) = |P_i|\psi\rangle|^2 = (P_i|\psi\rangle)^\dagger (P_i|\psi\rangle) = \langle\psi|P_i^2|\psi\rangle = \langle\psi|P_i|\psi\rangle \quad (6.14)$$

$$\text{Pr}(i) = \langle\psi|P_i|\psi\rangle = \text{Tr}(P_i|\psi\rangle\langle\psi|) \quad (6.19)$$

$$|\psi'\rangle = \frac{P_i|\psi\rangle}{\sqrt{\langle\psi|P_i|\psi\rangle}} \quad (6.20)$$

$$\langle A \rangle = \sum_i a_i \langle\psi|P_i|\psi\rangle \quad (6.21)$$

### Example

$$P_0 = |0\rangle\langle 0|$$

$$P_1 = |1\rangle\langle 1|$$

a) Express following projection operators in terms of  $P_0$  and  $P_1$

$$P_+ = |+\rangle\langle +|$$

$$P_- = |-\rangle\langle -|$$

b) Find

$$P_+ + P_- = ?$$

$$P_+ - P_- = ?$$

## Example 6.1

A system is in the state

$$|\psi\rangle = \frac{2}{\sqrt{19}}|u_1\rangle + \frac{2}{\sqrt{19}}|u_2\rangle + \frac{1}{\sqrt{19}}|u_3\rangle + \frac{2}{\sqrt{19}}|u_4\rangle + \sqrt{\frac{6}{19}}|u_5\rangle$$

Where  $|u_i\rangle$  are orthonormal basis

a) Write down all possible orthonormal the projection operator

b) If  $H|n\rangle = n\varepsilon|n\rangle$

Find  $\langle H \rangle$

## Bell States

$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$|\beta_{01}\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}$$

$$|\beta_{10}\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

$$|\beta_{11}\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$$

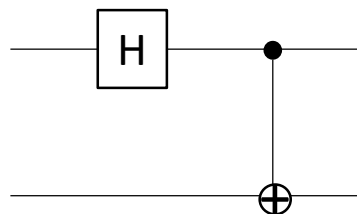
$$(7.27) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

$$(7.28) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

$$(7.29) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix}$$

$$(7.30) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

Circuit to create all above Bell States  
(with specific combinations of inputs)



Not all states  $|\psi\rangle \in H_A \otimes H_B$

$$|\psi\rangle = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

The state is **separable**  
if and only if

$$ad = bc$$

## Example 7.4

Alice and Bob each possess one member of a pair of interacting magnetic dipoles (spin-1/2 particles). The interaction Hamiltonian for two interacting magnetic dipoles separated by a distance  $r$  is given by

$$H_I = \frac{\mu^2}{r^3} (\vec{\sigma}_A \cdot \vec{\sigma}_B - 3Z_A Z_B) \quad \text{where } \vec{\sigma}_A = X_A \hat{x} + Y_A \hat{y} + Z_A \hat{z} \text{ and similarly for Bob.}$$

- a) Express  $H_I$  in matrix form
- b) Show that  $|\beta_{01}\rangle, |\beta_{11}\rangle, |00\rangle$  and  $|11\rangle$  are eigen states of  $H_I$
- c) Express  $H_I$  in the form of eigen vectors (in matrix form  $H_I$ )

### Solution

$$\vec{\sigma}_A \cdot \vec{\sigma}_B = X_A \otimes X_B + Y_A \otimes Y_B + Z_A \otimes Z_B$$

Substituting matrix form  
of Pauli operators

$$H_I = \frac{\mu^2}{r^3} \begin{pmatrix} -2 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & -2 \end{pmatrix}$$



b) Show that  $|\beta_{01}\rangle, |\beta_{11}\rangle, |00\rangle$  and  $|11\rangle$  are eigen states of  $H_I$

$$H_I|\beta_{01}\rangle = C_1|\beta_{01}\rangle$$

$$H_I|00\rangle = C_2|00\rangle$$

$$H_I|11\rangle = C_3|11\rangle$$

$$H_I|\beta_{11}\rangle = C_4|\beta_{11}\rangle$$

c) Express  $H_I$  in the form of eigen vectors (in matrix form)

$$H_I = \begin{pmatrix} C_1 & & & \\ & C_2 & & \\ & & C_3 & \\ & & & C_4 \end{pmatrix}$$

## USING BELL STATES FOR DENSITY OPERATOR REPRESENTATION

A density operator for a two-qubit system that is diagonal with respect to the Bell basis can be represented in terms of the Bell states using the expansion

$$\begin{aligned}\rho &= \sum_{i,j} c_{ij} |\beta_{ij}\rangle \langle \beta_{ij}| \\ &= c_{00} |\beta_{00}\rangle \langle \beta_{00}| + c_{01} |\beta_{01}\rangle \langle \beta_{01}| + c_{10} |\beta_{10}\rangle \langle \beta_{10}| + c_{11} |\beta_{11}\rangle \langle \beta_{11}|\end{aligned}\quad (7.38)$$

$\rho$  is separable if and only if

$$c_{00} \leq \frac{1}{2}$$

$$|\beta_{00}\rangle \langle \beta_{00}| = \frac{1}{4}(I \otimes I + X \otimes X - Y \otimes Y + Z \otimes Z) \quad (7.39)$$

$$|\beta_{01}\rangle \langle \beta_{01}| = \frac{1}{4}(I \otimes I + X \otimes X + Y \otimes Y - Z \otimes Z) \quad (7.40)$$

$$|\beta_{10}\rangle \langle \beta_{10}| = \frac{1}{4}(I \otimes I - X \otimes X + Y \otimes Y + Z \otimes Z) \quad (7.41)$$

$$|\beta_{11}\rangle \langle \beta_{11}| = \frac{1}{4}(I \otimes I - X \otimes X - Y \otimes Y - Z \otimes Z) \quad (7.42)$$

### Example 7.7

A density matrix for a certain two-qubit system in the  $\{ |00\rangle, |01\rangle, |10\rangle, |11\rangle \}$  basis is

$$\rho = \begin{pmatrix} \frac{1}{8} & 0 & 0 & \frac{1}{8} \\ 0 & \frac{3}{8} & \frac{-3}{8} & 0 \\ 0 & \frac{-3}{8} & \frac{3}{8} & 0 \\ \frac{1}{8} & 0 & 0 & \frac{1}{8} \end{pmatrix}$$

Can this state be written in a diagonal form with respect to the Bell basis? Is this a separable state?

#### Solution

Using the location of each matrix element  $\rho$  can be expressed in ket-bra notation

$$\begin{aligned} \rho = & \frac{1}{8}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|) \\ & + \frac{3}{8}(|01\rangle\langle 01| - |01\rangle\langle 10| - |10\rangle\langle 01| + |10\rangle\langle 10|) \end{aligned}$$



$$\rho = \frac{1}{4}|\beta_{00}\rangle\langle\beta_{00}| + \frac{3}{4}|\beta_{11}\rangle\langle\beta_{11}|$$

Since  $c_{00} = 1/4 < 1/2$ ,



this is a separable state.

## SCHMIDT DECOMPOSITION

### Consider $|\psi\rangle$ of two qubit system

Consider a composite system  $H_A \otimes H_B$ , and let  $|\psi\rangle \in H_A \otimes H_B$  be a pure state. Then there exists an expansion of  $|\psi\rangle$  the form

$$|\psi\rangle = \sum_i \lambda_i |a_i\rangle |b_i\rangle \quad (7.44)$$

where  $|a_i\rangle$  are orthonormal states belonging to system A and  $|b_i\rangle$  are orthonormal states belonging to system B (these states are called the **Schmidt bases for system A and system B**, respectively).

The expansion coefficients  $\lambda_i$  are such that  $\lambda_i \geq 0$  and  $\sum_i \lambda_i = 1$ . We call the  $\lambda_i$  **Schmidt coefficients** and the expansion (7.44) is known as the Schmidt decomposition.

The Schmidt coefficients are calculated from the matrix

$$Tr_B (|\psi\rangle\langle\psi|) \quad (7.45)$$

The **Schmidt number** is the number of nonzero eigenvalues  $\lambda_i$ . The Schmidt number is used in the following way:

- If a **state is separable**, then the **Schmidt number is 1**.
- If a **state is entangled**, then the **Schmidt number is > 1**.

We will denote the Schmidt number by Sch

### Example 7.8

Consider the state  $|\psi\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle)$ . Is this state separable? What is the Schmidt number?

#### Solution

$$\begin{aligned}\rho = |\psi\rangle\langle\psi| &= \frac{1}{4}(\underbrace{|00\rangle\langle 00|}_{\text{green}} - \underbrace{|00\rangle\langle 01|}_{\text{green}} - \underbrace{|00\rangle\langle 10|}_{\text{green}} + \underbrace{|00\rangle\langle 11|}_{\text{green}} - \underbrace{|01\rangle\langle 00|}_{\text{green}} + \underbrace{|01\rangle\langle 01|}_{\text{green}} \\ &\quad + \underbrace{|01\rangle\langle 10|}_{\text{green}} - \underbrace{|01\rangle\langle 11|}_{\text{green}} - \underbrace{|10\rangle\langle 00|}_{\text{green}} + \underbrace{|10\rangle\langle 01|}_{\text{green}} + \underbrace{|10\rangle\langle 10|}_{\text{green}} - \underbrace{|10\rangle\langle 11|}_{\text{green}} + \underbrace{|11\rangle\langle 00|}_{\text{green}} \\ &\quad - \underbrace{|11\rangle\langle 01|}_{\text{green}} - \underbrace{|11\rangle\langle 10|}_{\text{green}} + \underbrace{|11\rangle\langle 11|}_{\text{green}})\end{aligned}$$

$$\rho_A = \text{Tr}_B(|\psi\rangle\langle\psi|) = \langle 0|\psi\rangle\langle\psi|0\rangle + \langle 1|\psi\rangle\langle\psi|1\rangle$$

$$\rho_A = \text{Tr}_B(\rho) = \frac{1}{4}(|0\rangle\langle 0| - |0\rangle\langle 1| - |1\rangle\langle 0| + |1\rangle\langle 1|) + \frac{1}{4}(|0\rangle\langle 0| - |0\rangle\langle 1| - |1\rangle\langle 0| + |1\rangle\langle 1|)$$

$$\rho_A = \text{Tr}_B(\rho_{AB})$$

$$= \frac{1}{2}(|0\rangle\langle 0| - |0\rangle\langle 1| - |1\rangle\langle 0| + |1\rangle\langle 1|)$$

$$\rho_A = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad \text{With eigen values } \lambda_1 = 1 \text{ and } \lambda_2 = 0$$



Sch = 1

The Schmidt number is the number of nonzero eigenvalues



this is a separable state

$$|\psi\rangle = \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right)$$

### Example 7.8

Consider a singlet state  $|S\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}$

Show that it is an entangled state by computing Schmidt number

#### Solution

$$\begin{aligned}\rho &= |S\rangle\langle S| = \left(\frac{|01\rangle - |10\rangle}{\sqrt{2}}\right) \left(\frac{\langle 01| - \langle 10|}{\sqrt{2}}\right) \\ &= \frac{1}{2}(|01\rangle\langle 01| - |01\rangle\langle 10| - |10\rangle\langle 01| + |10\rangle\langle 10|)\end{aligned}$$

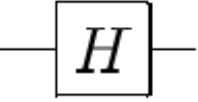
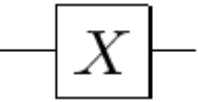
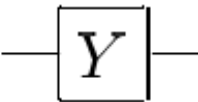
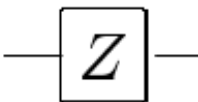
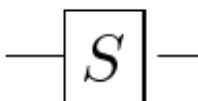
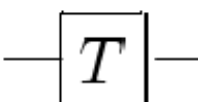
Taking the trace of B

$$\begin{aligned}\rho_A &= Tr_B \left( \frac{1}{2}(|01\rangle\langle 01| - \underline{|01\rangle\langle 10|} - \underline{|10\rangle\langle 01|} + |10\rangle\langle 10|) \right) \\ &= \frac{1}{2}(|1\rangle\langle 1| + |0\rangle\langle 0|)\end{aligned}$$

# Quantum Gates

Amazingly, Unitary is the only constraint on quantum gates

**Frequently used quantum gates and the circuit symbol**  
and their operator representation in matrix form

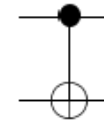
|            |                                                                                       |                                                                    |
|------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Hadamard   |    | $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ |
| Pauli- $X$ |    | $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$                     |
| Pauli- $Y$ |    | $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$                    |
| Pauli- $Z$ |    | $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$                    |
| Phase      |  | $\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$                     |
| $\pi/8$    |  | $\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$            |

# Quantum Gates

Amazingly, Unitary is the only constraint on quantum gates

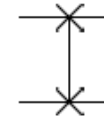
**Frequently used quantum gates and the circuit symbol** and their operator representation in matrix form

controlled-NOT



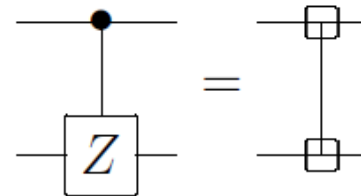
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

swap



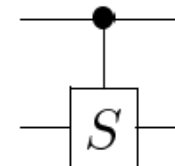
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

controlled-Z



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

controlled-phase



$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & i \end{bmatrix}$$



# Quantum Gates

Amazingly, Unitary is the only constraint on quantum gates

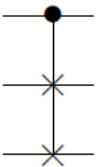
Frequently used quantum gates and the circuit symbol and their operator representation in matrix form

Toffoli



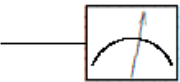
$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Fredkin (controlled-swap)



$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

measurement



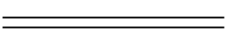
Projection onto  $|0\rangle$  and  $|1\rangle$

qubit



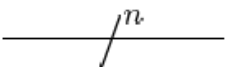
wire carrying a single qubit  
(time goes left to right)

classical bit



wire carrying a single classical bit

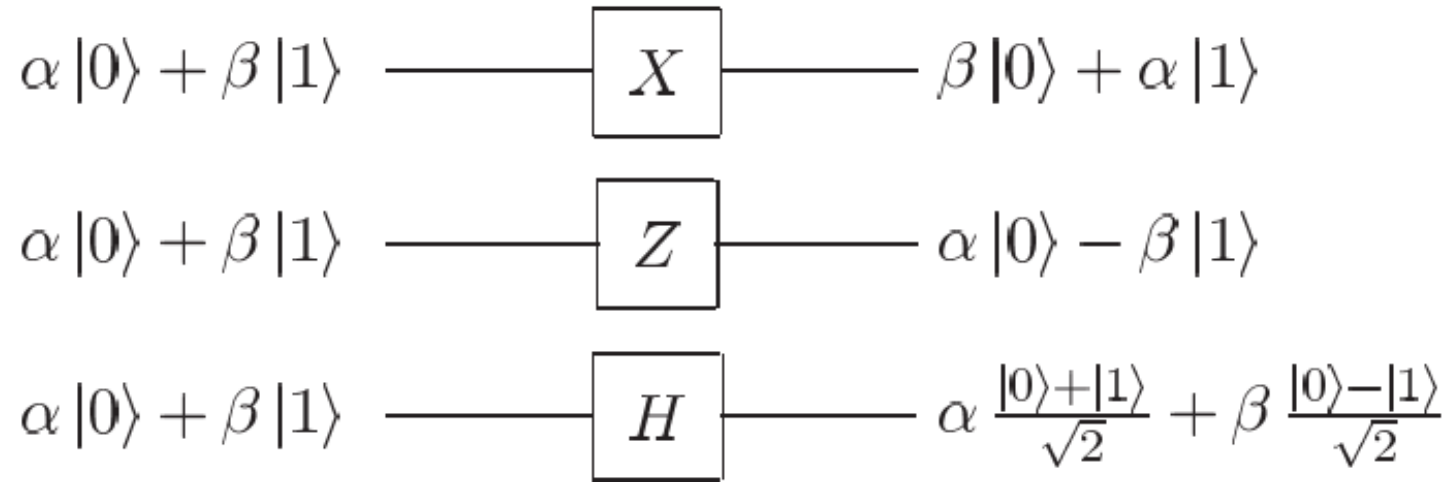
$n$  qubits



wire carrying  $n$  qubits

# Single Qubit Circuits

Amazingly, Unitary is the only constraint on quantum gates

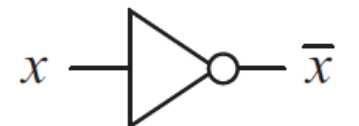


$$X|+\rangle = ?$$

$$H|+\rangle = ?$$

$$Z|-\rangle = ?$$

Only single bit classical circuit



# CNOT gate

$$|AB\rangle \rightarrow |A(B \oplus A)\rangle$$

$$CNOT = U_c = |00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 11| + |11\rangle\langle 01|$$

$$|0\rangle\langle 0| = \frac{1}{2}(I + \sigma_z)$$

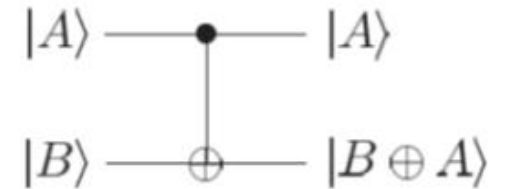
$$|1\rangle\langle 1| = \frac{1}{2}(I - \sigma_z)$$

$$|1\rangle\langle 0| = \frac{1}{2}(\sigma_x - i\sigma_y)$$

$$|0\rangle\langle 1| = \frac{1}{2}(\sigma_x + i\sigma_y)$$

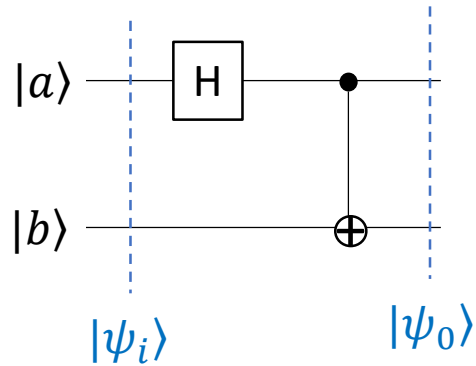
$$CNOT = U_c = \frac{1}{2} I \otimes I + \frac{1}{2} \sigma_z \otimes I + \frac{1}{2} I \otimes \sigma_x - \frac{1}{2} \sigma_z \otimes \sigma_x$$

controlled-NOT



$$U_{CN} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

# Circuit to Generate Entangled states

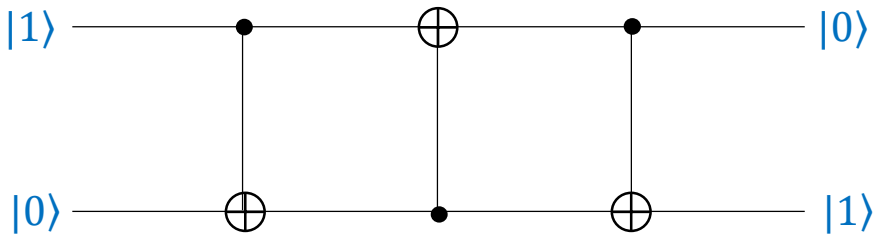
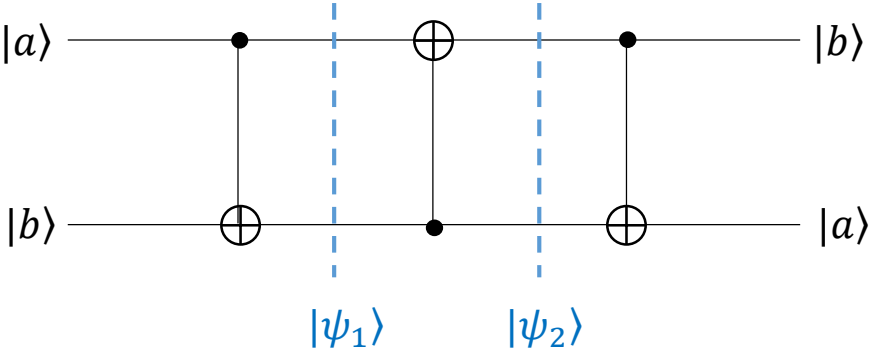


$$|\beta_{ab}\rangle = \frac{|0, b\rangle + (-1)^a |1, \bar{a}\rangle}{\sqrt{2}}$$

$$|\psi_i\rangle = |ab\rangle$$

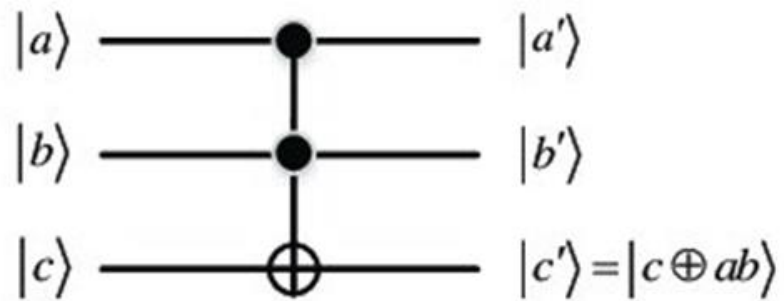
$$|\psi_o\rangle = |\beta_{ab}\rangle = \frac{1}{\sqrt{2}} [|0b\rangle + (-1)^1 |1\bar{a}\rangle]$$

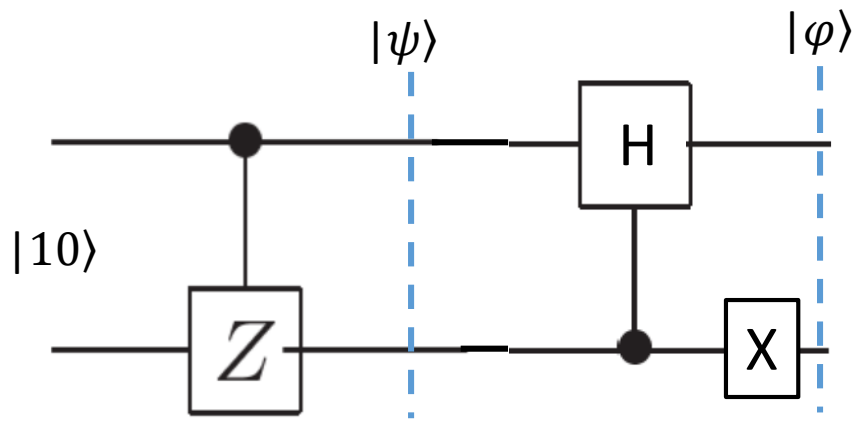
# Bit Flip Circuit



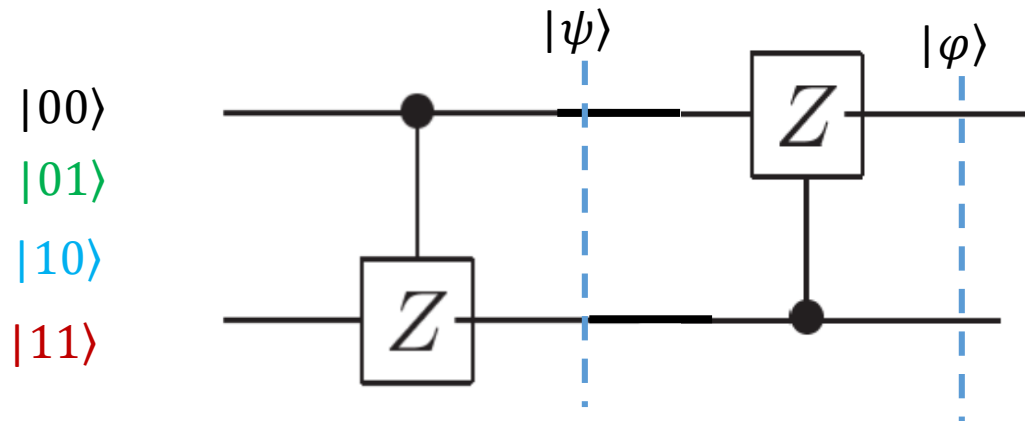
# Toffoli gate

| Inputs |     |     | Outputs |      |      |
|--------|-----|-----|---------|------|------|
| $a$    | $b$ | $c$ | $a'$    | $b'$ | $c'$ |
| 0      | 0   | 0   | 0       | 0    | 0    |
| 0      | 0   | 1   | 0       | 0    | 1    |
| 0      | 1   | 0   | 0       | 1    | 0    |
| 0      | 1   | 1   | 0       | 1    | 1    |
| 1      | 0   | 0   | 1       | 0    | 0    |
| 1      | 0   | 1   | 1       | 0    | 1    |
| 1      | 1   | 0   | 1       | 1    | 1    |
| 1      | 1   | 1   | 1       | 1    | 0    |



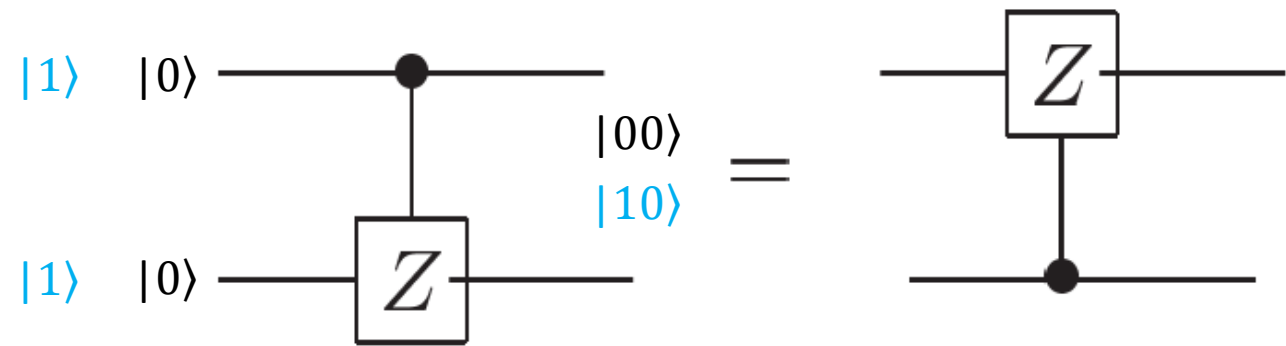


Find  $|\psi\rangle$  and  $|\varphi\rangle$  for given input

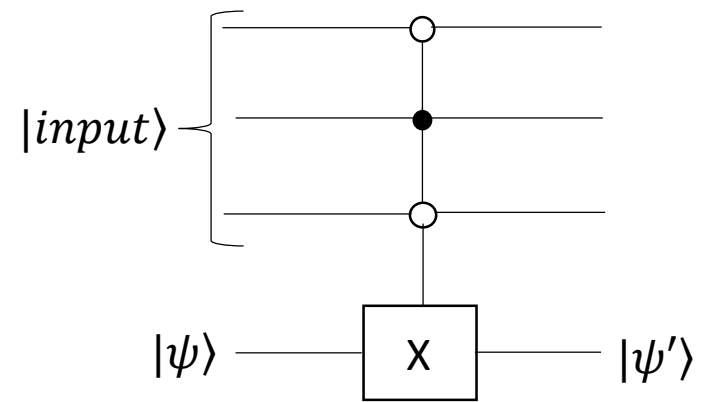


Find  $|\psi\rangle$  and  $|\varphi\rangle$  for all possible inputs

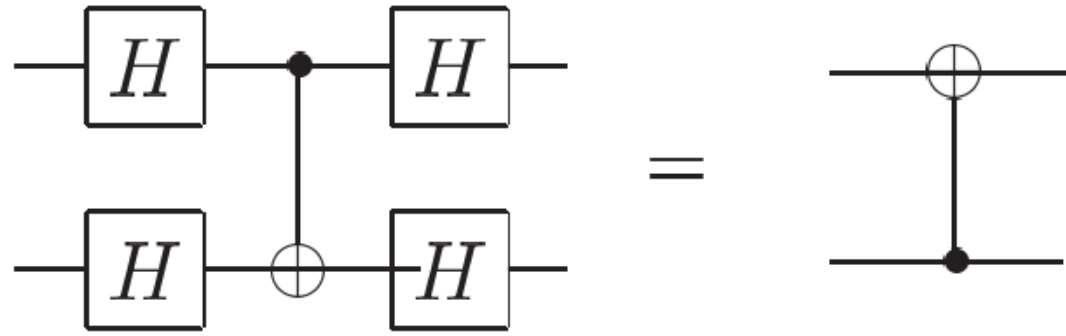
Exercise 4.18: Show that







CNOT control operation with  $|010\rangle$  state



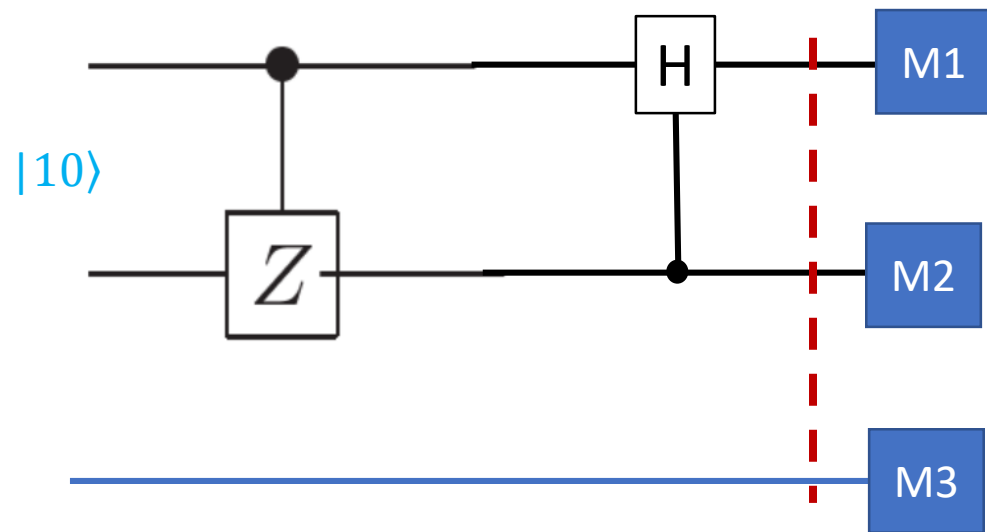
$$|+\rangle|+\rangle \rightarrow |+\rangle|+\rangle \quad (4.24)$$

$$|-\rangle|+\rangle \rightarrow |-\rangle|+\rangle \quad (4.25)$$

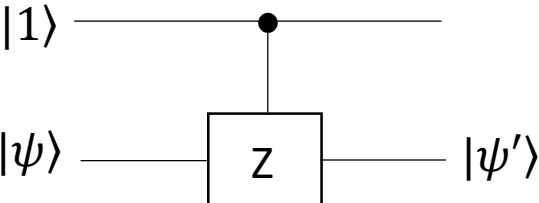
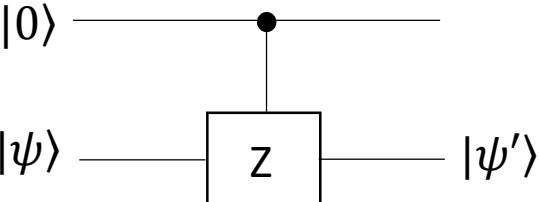
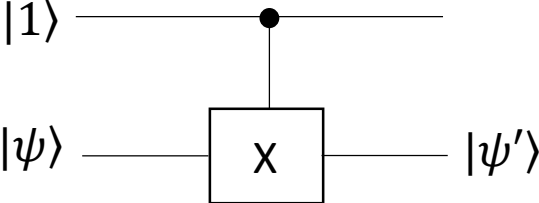
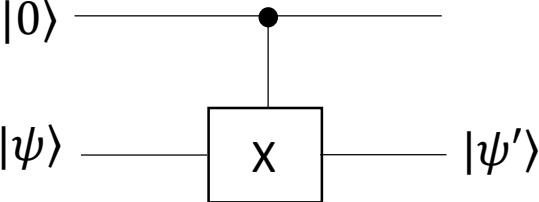
$$|+\rangle|-\rangle \rightarrow |-\rangle|-\rangle \quad (4.26)$$

$$|-\rangle|-\rangle \rightarrow |+\rangle|-\rangle. \quad (4.27)$$

Thus, with respect to this new basis, the state of the target qubit is not changed, while the state of the control qubit is flipped if the target starts as  $|-\rangle$ , otherwise

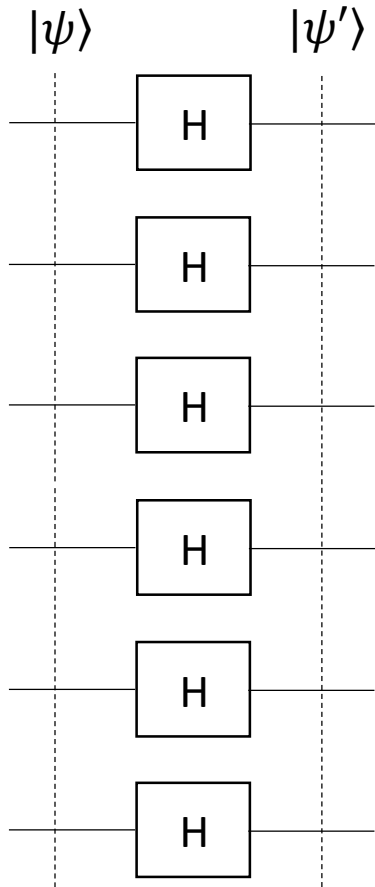


For given input  $|\psi\rangle$  calculate  $|\psi'\rangle$



$$|\psi\rangle = |000000\rangle$$

$$|\psi'\rangle = ?$$



$$|\psi\rangle = |111111\rangle$$

$$|\psi'\rangle = ?$$

# Quantum Teleportation

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\psi_1\rangle = |\psi\rangle|\beta_{00}\rangle$$

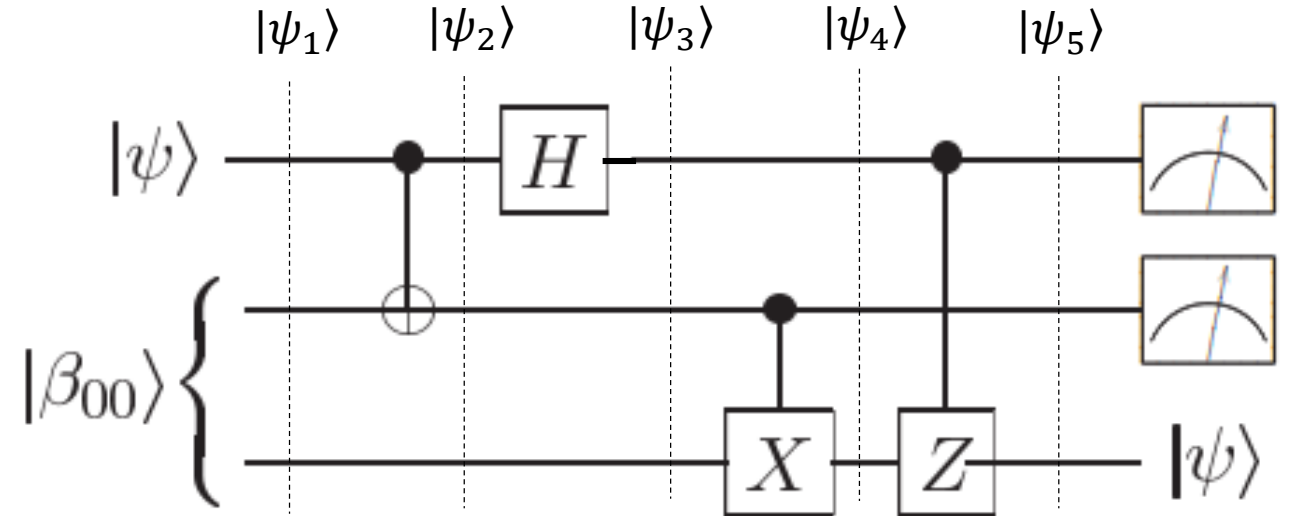
$$|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} [\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|00\rangle + |11\rangle)]$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} [\alpha|0\rangle(|00\rangle + |11\rangle) + \beta|1\rangle(|10\rangle + |01\rangle)]$$

$$|\psi_3\rangle = \frac{1}{\sqrt{2}} \left[ \alpha \left( \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \right) (|00\rangle + |11\rangle) + \beta \left( \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \right) (|10\rangle + |01\rangle) \right]$$

$$|\psi_3\rangle = \frac{1}{2} [\alpha|0\rangle|00\rangle + \alpha|0\rangle|11\rangle + \alpha|1\rangle|00\rangle + \alpha|1\rangle|11\rangle + \beta|0\rangle|10\rangle + \beta|0\rangle|01\rangle - \beta|1\rangle|10\rangle - \beta|1\rangle|01\rangle]$$



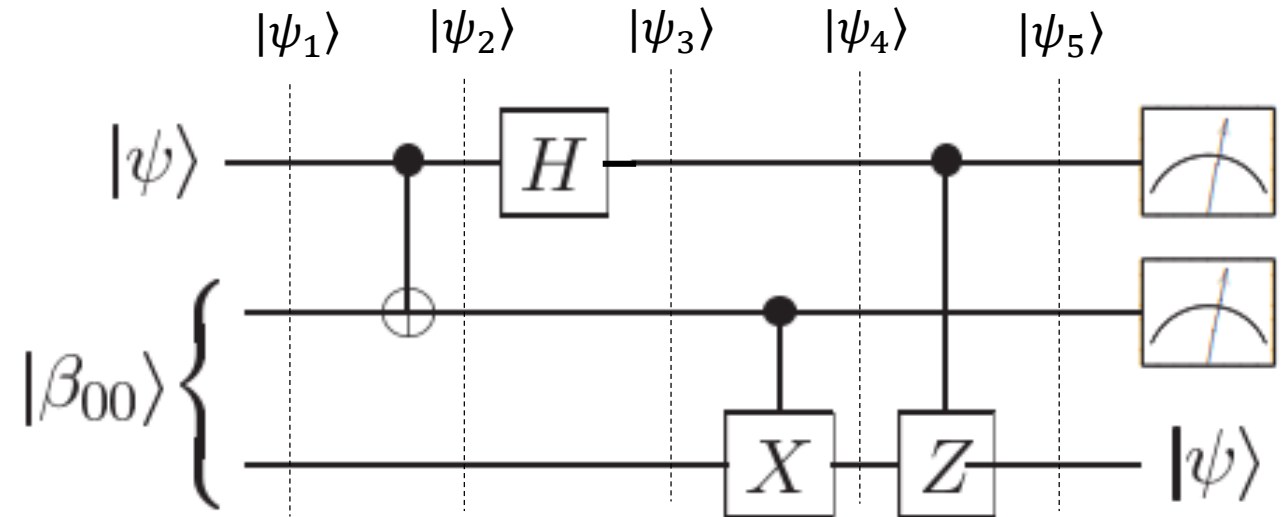
$$|\psi_3\rangle = \frac{1}{2} [\alpha \underbrace{|0\rangle|00\rangle}_{\text{green}} + \alpha \underbrace{|0\rangle|11\rangle}_{\text{purple}} + \alpha \underbrace{|1\rangle|00\rangle}_{\text{yellow}} + \alpha \underbrace{|1\rangle|11\rangle}_{\text{purple}} + \beta \underbrace{|0\rangle|10\rangle}_{\text{purple}} + \beta \underbrace{|0\rangle|01\rangle}_{\text{green}} - \beta \underbrace{|1\rangle|10\rangle}_{\text{yellow}} - \beta \underbrace{|1\rangle|01\rangle}_{\text{yellow}}]$$

$$|\psi_3\rangle = \frac{1}{2} [|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)]$$

Alice M

↓

$$\begin{aligned} 00 &\mapsto |\psi_3(00)\rangle \equiv [\alpha|0\rangle + \beta|1\rangle] \\ 01 &\mapsto |\psi_3(01)\rangle \equiv [\alpha|1\rangle + \beta|0\rangle] \\ 10 &\mapsto |\psi_3(10)\rangle \equiv [\alpha|0\rangle - \beta|1\rangle] \\ 11 &\mapsto |\psi_3(11)\rangle \equiv [\alpha|1\rangle - \beta|0\rangle] \end{aligned}$$



# Entanglement Swapping

In an interesting extension of teleportation, we can cause two particles that have never interacted to become entangled. This is possible, in principle, even if the particles are light years apart

Entanglement swapping begins with two EPR pairs. We label the qubits 1, 2, 3, and 4. **Alice has qubits 1 and 4** in her possession while **Bob has qubits 2 and 3** in his possession. Qubits 1 and 2 are entangled in the Bell state

**Qubits 1 and 2 are entangled in the Bell state**

$$|\beta_{00}\rangle_{12} = \frac{|00\rangle_{12} + |11\rangle_{12}}{\sqrt{2}}$$

**Qubits 3 and 4 are entangled in the Bell state**

$$|\beta_{00}\rangle_{34} = \frac{|00\rangle_{34} + |11\rangle_{34}}{\sqrt{2}}$$

**The product of these two states is**

$$\begin{aligned} |\beta_{00}\rangle_{12} |\beta_{00}\rangle_{34} &= \left( \frac{|00\rangle_{12} + |11\rangle_{12}}{\sqrt{2}} \right) \left( \frac{|00\rangle_{34} + |11\rangle_{34}}{\sqrt{2}} \right) \\ &= \frac{1}{2} (|00\rangle_{12} |00\rangle_{34} + |00\rangle_{12} |11\rangle_{34} + |11\rangle_{12} |00\rangle_{34} + |11\rangle_{12} |11\rangle_{34}) \end{aligned}$$



$$|\beta_{00}\rangle_{12}|\beta_{00}\rangle_{34} = \frac{1}{2}(|00\rangle_{12}|00\rangle_{34} + |00\rangle_{12}|11\rangle_{34} + |11\rangle_{12}|00\rangle_{34} + |11\rangle_{12}|11\rangle_{34})$$

Now let's do some simple algebra. First we just rearrange the qubits in each term so that we can write **qubits 1 and 4 together** and **qubits 2 and 3 together**:

$$|\beta_{00}\rangle_{12}|\beta_{00}\rangle_{34} = \frac{1}{2}(|00\rangle_{14}|00\rangle_{23} + |01\rangle_{14}|01\rangle_{23} + |10\rangle_{14}|10\rangle_{23} + |11\rangle_{12}|11\rangle_{34})$$

Now consider

$$\begin{aligned} |\beta_{00}\rangle_{14}|\beta_{00}\rangle_{23} &= \left( \frac{|00\rangle_{14} + |11\rangle_{14}}{\sqrt{2}} \right) \left( \frac{|00\rangle_{23} + |11\rangle_{23}}{\sqrt{2}} \right) \\ &= \frac{1}{2}(|00\rangle_{14}|00\rangle_{23} + |00\rangle_{14}|11\rangle_{23} + |11\rangle_{14}|00\rangle_{23} + |11\rangle_{14}|11\rangle_{23}) \end{aligned}$$

These two  
are not same



But can be, again consider

$$\begin{aligned} |\beta_{00}\rangle_{12}|\beta_{00}\rangle_{34} &= \frac{1}{2}(|00\rangle_{14}|00\rangle_{23} + |01\rangle_{14}|01\rangle_{23} + |10\rangle_{14}|10\rangle_{23} + |11\rangle_{12}|11\rangle_{34} \\ &\quad + \underline{|00\rangle_{14}|11\rangle_{23}} + \underline{|11\rangle_{14}|00\rangle_{23}} - \underline{|00\rangle_{14}|11\rangle_{23}} - \underline{|11\rangle_{14}|00\rangle_{23}}) \end{aligned}$$

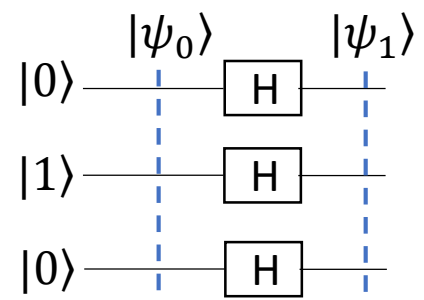
$$|\beta_{00}\rangle_{12}|\beta_{00}\rangle_{34} = \frac{1}{2}(|\beta_{00}\rangle_{14}|\beta_{00}\rangle_{23} + |01\rangle_{14}|01\rangle_{23} + |10\rangle_{14}|10\rangle_{23} \\ - |00\rangle_{14}|11\rangle_{23} - |11\rangle_{14}|00\rangle_{23})$$

Similar algebraic tricks on the other terms can be used to show that in the end

$$|\beta_{00}\rangle_{12}|\beta_{00}\rangle_{34} = \frac{1}{2}(|\beta_{00}\rangle_{14}|\beta_{00}\rangle_{23} + |\beta_{10}\rangle_{14}|\beta_{10}\rangle_{23} \\ + |\beta_{01}\rangle_{14}|\beta_{01}\rangle_{23} + |\beta_{11}\rangle_{14}|\beta_{11}\rangle_{23})$$

Remember, Alice has particles 1 and 4 in her possession. Now Alice performs a Bell state measurement on particles 1 and 4. The possible results are, of course,  $|\beta_{00}\rangle_{14}$ ,  $|\beta_{10}\rangle_{14}$ ,  $|\beta_{01}\rangle_{14}$ , and  $|\beta_{11}\rangle_{14}$ , each with a probability of  $\frac{1}{4}$ . Depending on the measurement result that Alice obtains, Bob's system collapses into one of the Bell state  $|\beta_{00}\rangle_{23}$ ,  $|\beta_{10}\rangle_{23}$ ,  $|\beta_{01}\rangle_{23}$ , or  $|\beta_{11}\rangle_{23}$ . Now particles 2 and 3 are entangled.

Let's summarize the procedure. Alice and Bob create two entangled states. Particles 1 and 2 are entangled together, and particles 3 and 4 are entangled together. Alice has particles 1 and 4 in her possession, and Bob has particles 2 and 3 in his possession. Bob carefully flies off to a distant land with his particles, and Alice does a Bell state measurement on her particles. The state collapses and Bob's particles 2 and 3 become entangled with each other. What is teleported here? Perhaps you can say that the entanglement has been teleported. This procedure works even if particles 2 and 3 have never interacted. To see the possible significance of this, add a twist and say that Charlie takes particle 3 and goes off somewhere else before the measurement is made. Alice does her Bell state measurement—now Bob and Charlie share an entangled pair and can use it as a resource between the two of them—so Bob can teleport a state to Charlie.



What is  $|\psi_0\rangle$  and  $|\psi_1\rangle$

$$(H \otimes H)|0\rangle|0\rangle = H^{\otimes 2} |0\rangle^{\otimes 2}$$

$$\Rightarrow H^{\otimes 2}|0\rangle^{\otimes 2} = \frac{1}{\sqrt{2^2}} \sum_{x \in \{0,1\}^2} |x\rangle \quad \Rightarrow \quad H^{\otimes 2}|0\rangle^{\otimes 2} = \frac{2}{\sqrt{2^2}} [|00\rangle + |01\rangle + |10\rangle + |11\rangle]$$

$$(H \otimes H \otimes H \otimes H)|0\rangle|0\rangle|0\rangle|0\rangle = H^{\otimes 4}|0\rangle^{\otimes 4}$$

$$\Rightarrow H^{\otimes 4}|0\rangle^{\otimes 4} = \frac{1}{\sqrt{2^4}} \sum_{x \in \{0,1\}^4} |x\rangle \quad \Rightarrow \quad H^{\otimes 4}|0\rangle^{\otimes 4} = \frac{2}{\sqrt{2^4}} [|0000\rangle + |0001\rangle + |0010\rangle + \dots |1111\rangle]$$

$$H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle$$

$$(H \otimes H)|0\rangle|0\rangle = H^{\otimes 2} |0\rangle^{\otimes 2}$$

$$\Rightarrow H^{\otimes 2}|0\rangle^{\otimes 2} = \frac{1}{\sqrt{2^2}} \sum_{x \in \{0,1\}^2} |x\rangle \quad \Rightarrow \quad H^{\otimes 2}|0\rangle^{\otimes 2} = \frac{2}{\sqrt{2^2}} [|00\rangle + |01\rangle + |10\rangle + |11\rangle]$$

$$(H \otimes H)|0\rangle|1\rangle = H^{\otimes 2}|01\rangle \quad \Rightarrow \quad H^{\otimes 2}|01\rangle = \frac{1}{\sqrt{2^2}} \sum_{x \in \{0,1\}^2} (-1)^x |x\rangle$$

$$(H \otimes H)|1\rangle|0\rangle = H^{\otimes 2}|10\rangle \quad \Rightarrow \quad H^{\otimes 2}|10\rangle = \frac{1}{\sqrt{2^2}} \sum_{x \in \{0,1\}^2} (-1)^x |x\rangle$$

$$(H \otimes H)|1\rangle|1\rangle = H^{\otimes 2}|11\rangle \quad \Rightarrow \quad H^{\otimes 2}|11\rangle = \frac{1}{\sqrt{2^2}} \sum_{x \in \{0,1\}^2} (-1)^{x_0 \oplus x_1} |x\rangle$$

Decimal value of x

## Quantum Inference

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$H|\psi\rangle = \left(\frac{\alpha + \beta}{\sqrt{2}}\right)|0\rangle + \left(\frac{\alpha - \beta}{\sqrt{2}}\right)|1\rangle$$

Therefore new probability amplitudes are  $\alpha \rightarrow \frac{\alpha + \beta}{\sqrt{2}}$  and  $\beta \rightarrow \frac{\alpha - \beta}{\sqrt{2}}$

If the initial state is

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \quad \Rightarrow \quad H|\psi\rangle = |0\rangle$$

**Quantum interference** plays an important role in the **development of quantum algorithms**:

- **Quantum interference allows us to gain information about a function  $f(x)$**  that depends on evaluating the function at many values of  $x$ . That is, interference allows us to deduce certain global properties of the function





























BackUp

# Commutation Algebra

Commutation  
relation

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

$$[\hat{A}, \hat{B}] = 0 \quad \hat{A}\hat{B} = \hat{B}\hat{A} \quad \hat{A} \text{ and } \hat{B} \text{ Commutes}$$

$$[\hat{A}, \hat{B}] \neq 0 \quad \hat{A}\hat{B} \neq \hat{B}\hat{A} \quad \text{In general, } \hat{A} \text{ and } \hat{B} \text{ do not commute}$$

Some properties related to commutation

$$[\hat{A}, \hat{B} + \hat{C}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}]$$

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$$

$$[\hat{A}, \hat{A}^\dagger] = \hat{A}\hat{A}^\dagger - \hat{A}^\dagger\hat{A} = 0 \quad \text{If an operator } \hat{A} \text{ is a normal operator}$$

$$[\hat{X}, \hat{p}] = i\hbar \hat{I} \quad \hat{X}\text{-position and } \hat{p}\text{-momentum **DO NOT** commute}$$

If two operators commutes, they  
posses a set of common eigenvector